PROPOSED CHANGE

Wire wait time in SABRE routing

Prefer SWAPs whose physical qubits become available earlier

A small scheduling signal in the existing routing score

$$H_{new}(s) = H_{old}(s) + \lambda \max(r_a, r_b)$$

$\lambda = 2.2 \times 10^{-5}$ distance units per estimated layer

Latest commit: Add configurable wire wait time scoring to SABRE

SABRE routing in one loop

π: logical-to-physical layout D: shortest-path distance

$D(\pi(u), \pi(v)) = 1$

A two-qubit gate is routable when its operands are adjacent.

1 → 2 → 3 → 4

Front layer F

Gates with satisfied dependencies. Route adjacent gates first.

Candidate SWAPs

Hardware edges touching a qubit in a blocked front-layer gate.

Score each SWAP

Distance, lookahead, optional decay, and now wire wait time.

Apply and repeat

Insert the best SWAP. Update π and route newly adjacent gates.

Layout search

Forward and reverse routing refine the initial layout.

Repeated trials explore alternatives. Routing selects the fewest SWAPs.

The score before this commit

F: front layer Eℓ: lookahead layer π_s: layout after candidate SWAP s

$$S_L(\pi) = \Sigma_{(u,v)\in L}\ D(\pi(u), \pi(v))$$

$$C(\pi) = \alpha\, S_F(\pi) + \Sigma_\ell\, \beta_\ell\, S_{E\ell}(\pi)$$

Without decay

$$H_{old}(s) = C(\pi_s) - C(\pi)$$

With decay

$$H_{old}(s) = d_s\, C(\pi_s)$$

$$d_s = \max(decay_a, decay_b)$$

Lower scores win. Epsilon ties use the seeded random generator.

Qiskit defaults: basic $\alpha = 1/|F|$. Lookahead / decay $\alpha = 1$, $\beta_1 = 0.5/N$.

The new score prices wire availability

$$H_{new}(s) = H_{old}(s) + \lambda \max(r_a, r_b)$$

s = (a,b) r: estimated physical-wire ready times, initially zero

Routed two-qubit SABRE node

$t = \max(r_a, r_b) + 1$

Committed routing SWAP

$t = \max(r_a, r_b) + 3$

$r_a = r_b = t$ Both updates set the two physical wires to the same finish time.

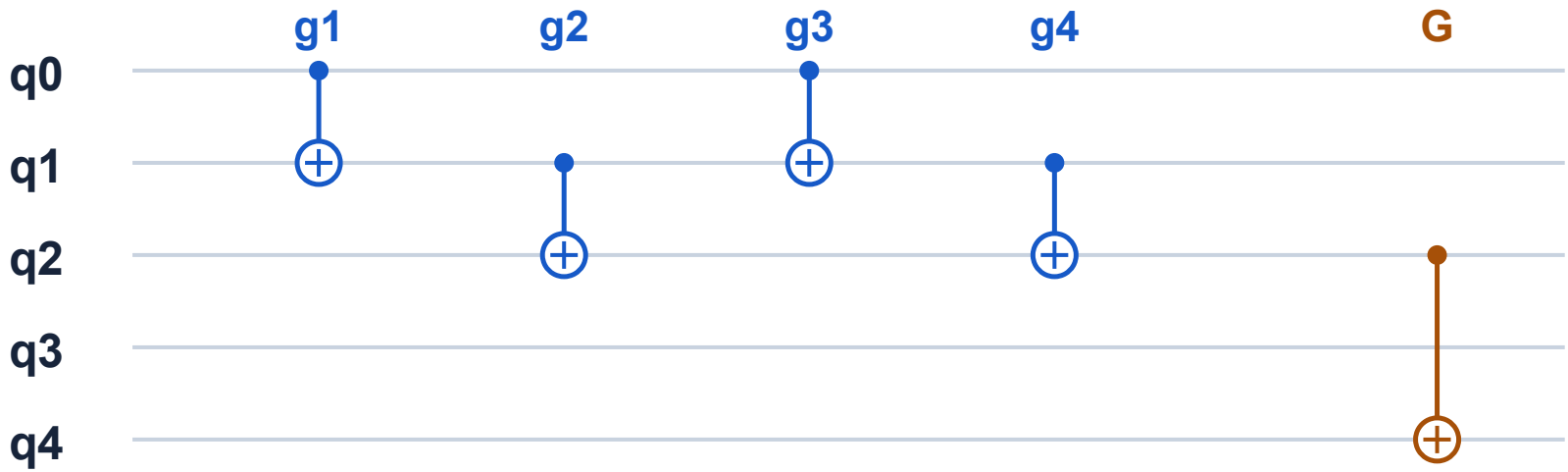
Added after decay The decay multiplier does not scale the wait penalty.

An estimated schedule One contracted node can contain several original gates.

One busy chain, two idle wires



q = logical qubit P = physical wire Initially, qi sits on Pi.



Gate to route: G

CX(q2,q4)

q2 is on P2.

q4 is on P4.

Distance = 2 edges

The blue gates share q1, so they run one after another. P3 and P4 remain idle.

After the prefix, P2 is busy and P3–P4 are free

Each CX sets both participating wires to max(their ready times) + 1.

After operation	P0	P1	P2	P3	P4
Start	0	0	0	0	0
g1: CX(P0,P1)	1	1	0	0	0
g2: CX(P1,P2)	1	2	2	0	0
g3: CX(P0,P1)	3	3	2	0	0
g4: CX(P1,P2)	**3**	**4**	**4**	**0**	**0**

A: SWAP(P2,P3)

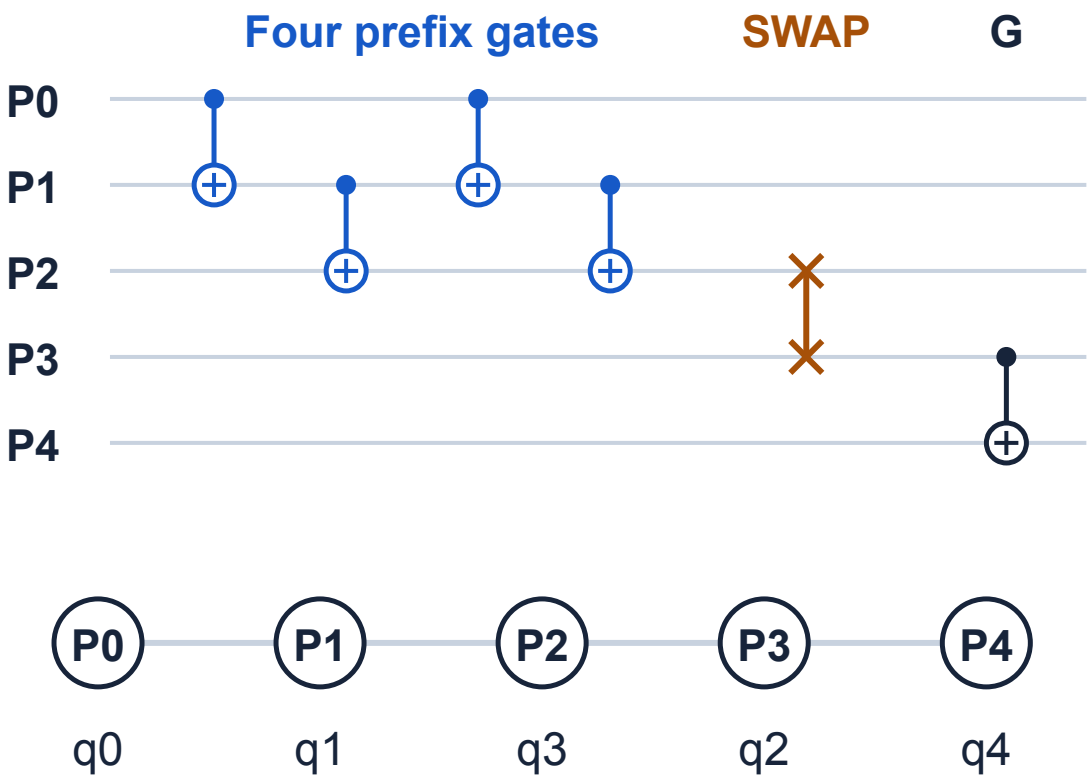
Earliest start = max(4,0) = 4

B: SWAP(P3,P4)

Earliest start = max(0,0) = 0

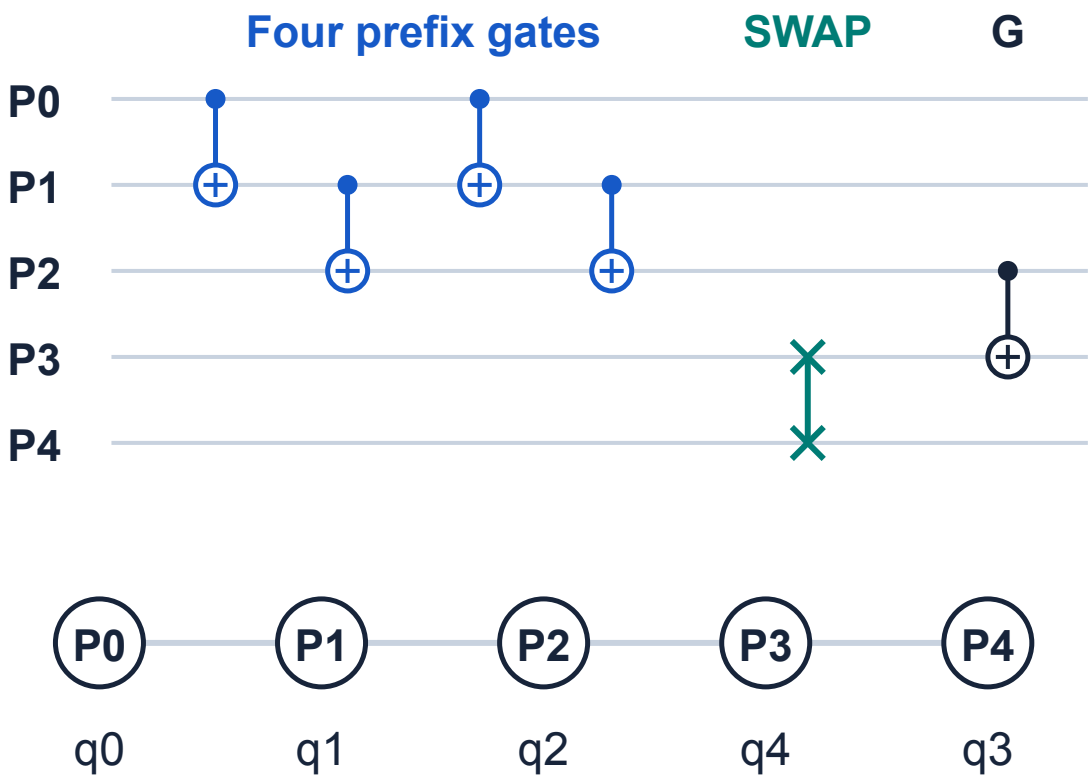
Both SWAPs make G executable

A: SWAP(P2,P3)



q2 moves to P3. G becomes CX(P3,P4).

B: SWAP(P3,P4)



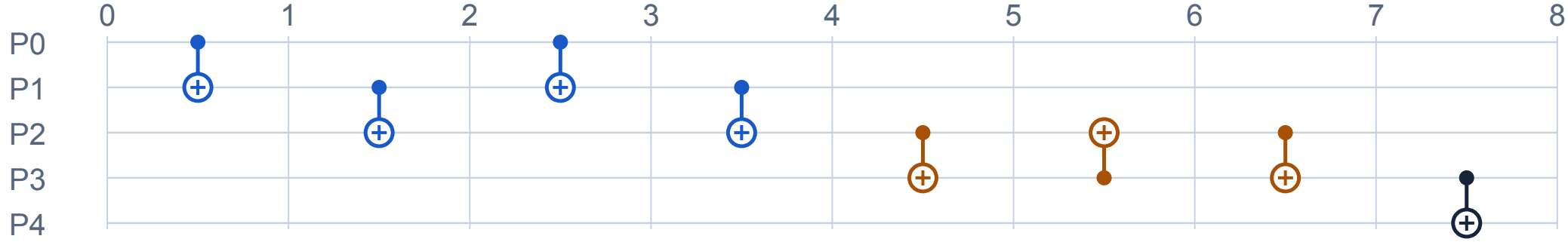
q4 moves to P3. G becomes CX(P2,P3).

One SWAP in each route, but 8 versus 5 layers

Each CX takes one layer. Each SWAP is three CX gates. No gate cancellation.

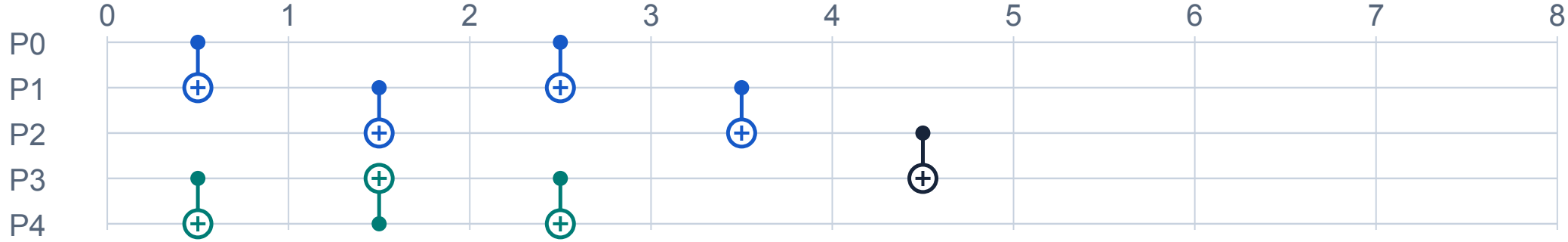
A: SWAP(P2,P3)

Finish at 8



B: SWAP(P3,P4)

Finish at 5



A: 4 + 3 + 1 = 8

B: max(4,3) + 1 = 5

Why the wait-time score chooses B

G starts at distance 2. Either SWAP reduces its distance to 1.

Quantity	A: SWAP(P2,P3)	B: SWAP(P3,P4)
Distance-score change	−1	−1
Earliest SWAP start	4	0
Added wait penalty	4λ = 0.000088	0
Total score	**−0.999912**	**−1.000000**

λ = 0.000022. Basic weight = 1. No lookahead or decay.

Without wait scoring: the two distance scores tie.

With wait scoring: B has the lower score and overlaps the busy prefix.